Storage Lab: Linux Application Volume Capacity Triage & Large File Discovery

Duration : 30 Min.

Scenario

An enterprise application server's dedicated data volume `/data` (or `/opt/app_data`) has reached 96% utilization with only 4% free space remaining. The administrator cannot delete files at random because mission-critical services depend on the volume. The engineer must log in via SSH to the Linux system, inspect filesystem mount states, execute a recursive search and sorting pipeline to identify the largest files (orphaned database dumps, stale staging tarballs, runaway debug logs), sort them in descending order of size, and generate a structured cleanup candidate report `large_files_report.csv`.

Task Objectives

Perform the following storage capacity triage tasks inside your `student_workspace` directory:

> **Pre-Staged Files in `student_workspace/`:** - **`mock_data_volume/`**: Simulated application storage hierarchy containing candidate files. - **`large_files_report.csv`**: Pre-staged CSV boilerplate file with standard headers (`FilePath`,`SizeMB`,`LastModified`).

1. Filesystem Capacity Inspection

- Inspect filesystem disk space usage to isolate mounted partitions and identify critical capacity exhaustion.

2. Recursive Search & Sorting Pipeline

- Recursively scan `mock_data_volume/` to discover all large candidate files and rank them in descending order by size.

3. Candidate Report Generation

- Directly edit `large_files_report.csv` in your workspace and replace the `TODO_*` placeholders with the top identified file paths, formatted sizes in MB (or GB), and modification timestamps.

Verification

Once you have populated `large_files_report.csv`, test your solution locally by running:

```
python run.py
```

Once verified, click **"Run Tests"** on the assessment platform to record your score.

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Volume query: accurate Linux filesystem inspection via df -hT isolating capacity bottlenecks	4 Marks
TC2	Recursive search & sorting: find command pipeline correctly identifies and sorts large files	3 Marks
TC3	CSV report formatting: large_files_report.csv contains valid headers, paths, formatted sizes, and timestamps	3 Marks

Total Score: 10 Marks

Cross-Platform Reference Note

In enterprise operations, Linux administrators execute recursive find/sort pipelines on storage mount points, whereas Windows administrators execute PowerShell cmdlets (`Get-Volume` and `Get-ChildItem`). Both SOPs fulfill the core change-controlled space recovery workflow.